
Detailed Design and Project Timeline

Lit Bits – Modular Display Devices

This is a very good report, earning 65.4 / 80. The overall design is strong and nicely justified. Section 3.7 is especially good.

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July 2, 2015

Table of Contents

1	Project Description	3
1.1	Motivation	3
1.2	Project Objective	3
1.3	Block Diagram	3
2	Project Specifications	5
2.1	Functional Specifications	5
2.2	Non-Functional Specifications	6
3	Detailed Design	7
3.1	User Interface	7
3.1.1	User Interface and Communication	7
3.1.2	Database	8
3.2	Animation Renderer	8
3.2.1	Requirements	8
3.2.2	Animation Renderer Implementation	9
3.2.3	Specifications	9
3.2.4	Alternatives	9
3.3	Addressing Algorithm	9
3.3.1	Requirements	9
3.3.2	Algorithm Implementation	9
3.3.3	Specifications	11
3.3.4	Alternatives	11
3.4	External Control Device Software	12
3.4.1	Requirements	12
3.4.2	External Control Device Software	12
3.4.3	Specifications	12
3.4.4	Alternatives	12
3.5	Mechanical Enclosure	13
3.5.1	Introduction	13
3.5.2	Requirements	13
3.5.3	Construction Material Selection	13
3.5.4	Enclosure Sizing	13
3.5.5	Mitigating Light Bleed	14
3.5.6	Light Diffusion	14
3.5.7	Equidistant Pixels	15
3.5.8	Proposed Design	16
3.6	Mating Connector Integration	16
3.6.1	Requirements	16
3.6.2	Ease of Use	17
3.6.3	Mating Direction	17
3.6.4	Current Capacity	18
3.6.5	Results	18
3.7	Lit Bit Module Microcontroller Software	18
3.7.1	Bus Interactions	18
3.7.2	Decoding Bus Packets for LED Use	20
3.8	LED Array Circuit	21
3.8.1	Requirements	21
3.8.2	Power Calculations	21
4	Section: Discussion and Project Timeline	23
4.1	Evaluation of Final Design	23

4.2	Use of Advanced Knowledge	23
4.3	Creativity, Novelty, and Elegance	24
4.4	Student Hours	24
4.5	Potential Safety Hazards	24
4.5.1	High Current	24
4.5.2	Eye Safety	24
4.5.3	Heat and Fire Safety	24
4.6	Project Timeline	25
5	Glossary	26
	References	28

1 Project Description

1.1 Motivation

Every year, 50 million people attend concert venues to see their favourite artists [1]. As part of these concert events, artists require, now more than ever, stimulating visuals and lighting effects to evoke emotions in their audience. Current modular display solutions are only affordable to successful artists due to their high cost per display unit along with the requirement of specialized animators and private contractors to maintain the functionality of these displays. The combined pricing structure of the equipment and personnel to operate these modular displays keeps them inaccessible to amateur or low-budget artists.

1.2 Project Objective

The aim of this project is to create an easy to use, functional, modular lighting system, capable of being used for a variety of lighting applications, where users can display seamless animations and invigorating visuals.

1.3 Block Diagram

The block diagram of the Lit Bits modular display system is shown in Figure 1.

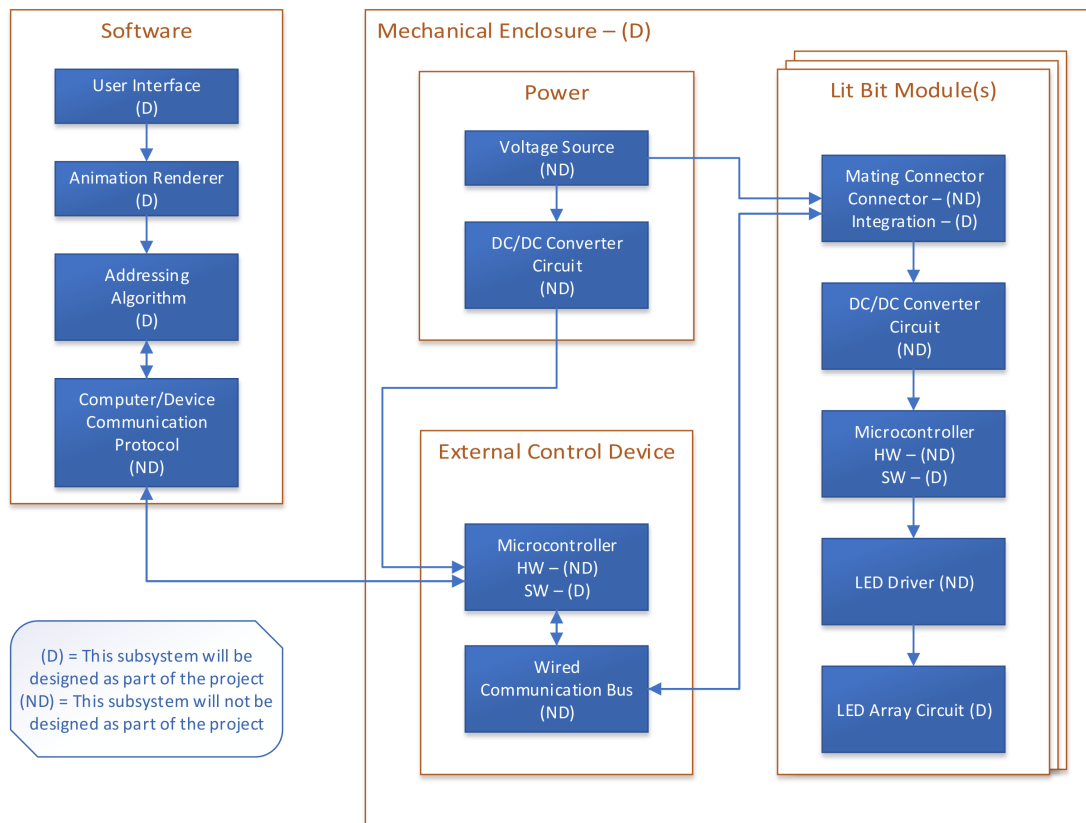


Figure 1: Lit Bits high-level block diagram.

The following is a comprehensive list of the different high-level components that make up this project.
(D – Designed, ND – Not Designed)

- **User Interface (D)**: Interface designed for users to interact with the Lit Bits to specify the animations to be displayed.
- **Animation Renderer (D)**: Reads the selected animation from the user interface and translates it into data which can be used by each Lit Bit module in order to display the animation.
- **Addressing Algorithm (D)**: This algorithm is used by the software to understand where each Lit Bit is with reference to the other Lit Bits, as well as their orientation.
- **Computer/Device Communication Protocol (ND)**: The physical medium in which data will be sent from the source (user input from the software) to the destination (external control device).
- **Mechanical Enclosure (D)**: The enclosure in which all electrical components are housed. This enclosure defines the look and feel of the Lit Bit modules and provides light isolation between pixels.
- **Mating Connector (Connector ND, Integration D)**: The power and data connector which enables power distribution amongst modules as well as inter-module communication over I2C.
- **External Control Device (Hardware ND, Software D)**: This device translates the user input from the software into the communication protocol that the Lit Bits are able to receive data from.
- **Wired Communication Bus (ND)**: The communication system that will traverse module specific animation frames to each individual Lit Bit from the external microcontroller.
- **Lit Bits Microcontroller (Hardware ND, Software D)**: This microcontroller will decompose data specified to it from the external microcontroller via the bus in order to determine what colours must be pushed to each individual pixel on the Lit Bit module.
- **Serial LED Driver (ND)**: The communication protocol necessary to address each pixel on the Lit Bit module. This serialized data contains colour and brightness data to be used by each pixel.
- **LED Array Circuit (D)**: The physical arrangement of the light-emitting diodes in the physical medium.

2 Project Specifications

2.1 Functional Specifications

The functional specifications of the project are detailed in Table 1.

Table 1: Functional specifications.

#	Functional Specification	Essential/Non-Essential	Complete?
1	Any string of Lit Bits in any configuration must act like a seamless display, regardless of its shape. (<i>Addressing Algorithm</i>)	Essential	
2	Lit Bit modules must be able to be connected to any of the three available expansion ports on another module to expand the display size in that direction. (<i>Mechanical Enclosure</i>)	Essential	
3	Any user should be able to connect two Lit Bit modules together in a physical manner in less than or equal to ten seconds. (<i>Mating Connector</i>)	Essential	
4	Any connected display of Lit Bit modules must be capable of a minimum of 24 frames per second (FPS) for the displayed animations. (<i>Animation Renderer, Computer/Device Communication Protocol, Wired Communication Bus</i>)	Essential	
5	The External Control Device must know the entire configuration of the Lit Bit network such that it can send subsets of visuals to each module. (<i>External Control Device, Addressing Algorithm</i>)	Non-Essential	
6	The User Interface must be able to relay settings selections to the External Control Device from the user. (<i>User Interface</i>)	Non-Essential	

2.2 Non-Functional Specifications

The non-functional specifications of the project are detailed in Table 2.

Table 2: Non-functional specifications.

#	Non-Functional Specification	Essential/Non-Essential	Complete?
1	Each module must be smaller than or equal to $25\text{ cm} \times 25\text{ cm} \times 6\text{ cm}$ (Length $\times$ Width $\times$ Height) (<i>Lit Bit Module(s)</i>)	Non-Essential	
2	Each module will feature an 12×12 matrix of rectangular cells that will properly disperse light from the LED in each cell across its own viewable surface. (<i>Lit Bit Module(s)</i>)	Essential	

3 Detailed Design

3.1 User Interface

The final design uses LAMP on the External Control Device (ECD). LAMP stands for Linux Apache MySQL PHP. These four components are used to run a web application on the ECD.

The user interface (UI) subsystem must have:

- A user interface.
- Communication between web application and server.
- Database for storing animations.

3.1.1 User Interface and Communication

Since a web application is used, any device that has internet access can be used to control the Lit Bit Modules (LBMs). When the web application is first loaded, it communicates with the MySQL database and gets a list of the animation names and creates a drop down control with the IDs and names of each animation. It communicates with the database by using PHP. Figure 2 is the PHP code that will access the MySQL database and get the ID and names from all of the rows in the animations table. It then creates an HTML drop down list and populates it with the ID and names of the animations stored in the table.

```
1 <?php
2 $db = mysql_connect('localhost', 'root', '');
3 mysql_select_db('litbits', $db);
4 $sql = 'SELECT ID,Name FROM animations';
5 $result = mysql_query($sql, $db);
6
7 echo "<select>";
8 while ($row = mysql_fetch_assoc($result)) {
9     echo "<option value=" . $row['ID'] .
10         ">" . $row['Name'] . "</option>";
11 }
12 echo "</select>";
13 ?>
```

Figure 2: PHP code to retrieve animation IDs and names to create a drop down list.

For the web application to communicate with the server after the user has selected an animation, there needs to be a web form transaction. A form is an HTML element that is used to get and send user input to a server. In this case, the user input is the users choice of an animation. When the user submits the form, it will send the users input to a PHP file which queries the database for animation data based on the ID of the animation the user selected. It then sends the data to the ECD which reads the data and displays the animation on the LBMs.

When the animation data is retrieved from the database, an SQL query is used to get the data. Since the query is based on user input, SQL injection protection must be implemented. *Prepared Statements* are used to prevent SQL injection [2]. Figure 3 shows the PHP code required to implement prepared statements.

```

1 $id = $_GET['id'];
2 $statement = $connection->prepare( "SELECT * FROM animations WHERE id = ?" );
3 $statement->bind_param( "i", $id );
4 $statement->execute();

```

Figure 3: Prepared statements used to prevent SQL injection during retrieval of animation data.

A prepared statement prevents SQL injection by escaping and using the user input as a parameter of an SQL query.

3.1.2 Database

The animations are saved in the MySQL database on the ECD. It is saved with a unique ID, animation name, and animation data. The ID is an auto-incrementing primary key for the animations. The animation name is a varchar database field with a length of 100 characters. The animation data is defined as raw text as the animation data can be very large and requires post-processing to interpret.

This design satisfies the specifications because it allows the user to change the animation that is displayed on the LBMs. When they make a selection on the web application, it relays the selection to the ECD which displays the animation on the LBMs.

An alternative design to storing the animations in the database is to have the animation data stored in pre-defined files on the ECD. The server uses PHP to read the animation data files, convert it to an object, and send it to the LBMs. The advantages of using files is that no extra software is needed, it is simple and easy to use, and takes up less disk space than information in a database. Although data files are a reasonable solution, a database is used as it can scale, is concurrent, and can fetch data quickly for the user. User accounts and animation builders are also good features to have for the web application, but they are not absolutely required.

The database, being concurrent, allows multiple users to use the database without causing problems. For files, multiple users could cause data corruption. Being able to query data allows for different kinds of information that would not be possible or too slow with files, such as displaying a long list of all stored animations. The final design focuses on providing implemented features to a single user.

3.2 Animation Renderer

3.2.1 Requirements

The Animation Renderer must meet the following requirements:

- Contain predefined animations.
- Contain a data structure to send animation to LBMs.
- Uses a UI to determine what to send.
- Map 24-bit colour to 16-bit colour.

3.2.2 Animation Renderer Implementation

Using predefined animations contained in array structures which have been generated, the animation renderer takes the inputs given from the UI and is able to make a request to the database to retrieve the animation data. The animation renderer is then able to package the data into an animation array which contains the RGB bit values corresponding to each LED on a LBM. The way it must package them is to take the existing 24-bit colour value which is generated from the input animation and using a 16-bit mapping algorithm, is able to downscale the colours so the data can be easily sent to the LBM. This effectively aids in reducing bandwidth consumption of the wired communication bus.

3.2.3 Specifications

The animation renderer meets the specifications as it is able to take the predefined animations and use the generated data to create multiple frames and store them into array data structures to be passed on to the ECD Software.

3.2.4 Alternatives

The alternatives to the animation renderer is that the animations could sit locally within each of the LBMs themselves. The downside to this alternative is that it requires more storage space at each LBM as well as more post-processing before the animation is displayed. The other alternative is to have the ECD send out the animations in a sequence, not needing any user input, and no extra processing is required. This solution would not work as the project requires the animations to be dynamic and selectable so that the user can choose what animations they want displayed across the LBMs.

3.3 Addressing Algorithm

3.3.1 Requirements

The Addressing Algorithm for a LBM must meet the following requirements:

- Must automatically assign an address to any new Lit Bit connected to the chain.
- Must record the following in an array on the ECD:
 - Assigned Address.
 - Connections relative to other Lit Bits.
 - Orientation to the connected Lit Bit.

3.3.2 Algorithm Implementation

The final design of the Addressing Algorithm is implemented on the I2C (Inter-Integrated Circuit) standard [3]. A new device that is connected to an already connected LBM automatically assumes the address of 0x7F (127) and notifies the adjacent device using the Serial Position Address Line (SPAL). The SPAL is a

separate serial line used to notify the adjacent LBM of which side the new module has been connected to. When a new device is connected, a processor interrupt starts a notification sequence on the adjacent LBM. The adjacent LBM waits for the I2C bus to be free then requests access to the bus. The slave LBM then notifies the master (ECD) that there is a new device on the bus (the ECD knows that new devices are at address 0x7F), which then checks the address array to find the next available address for a LBM and assigns that new LBM with said address. Along with the address of the new LBM, the master records the address of the adjacent LBM as well as the position of the new LBM with respect to the LBM it is connected to. Figure 4 below shows the different orientations a LBM can have.

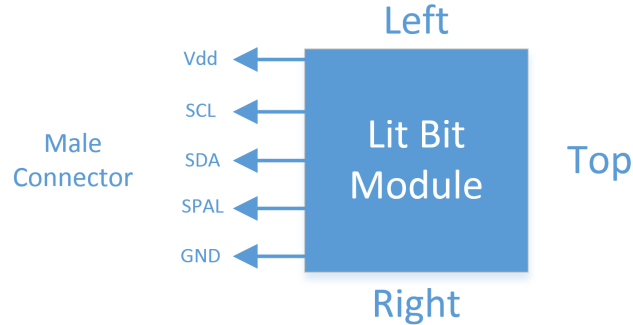


Figure 4: Lit Bit Module Orientation

The array found on the ECD looks similar to Figure 5 below.

Addr	Adjacent Addr	Orientation
0x01	0x00	Top
0x02	0x01	Left
0x03	0x00	Right
0x04	0x01	Top
...	...	...
0xNN	0xNN	X

Figure 5: External Control Device Lit Bit Module Array

This example array is interpreted as shown in Figure 6.

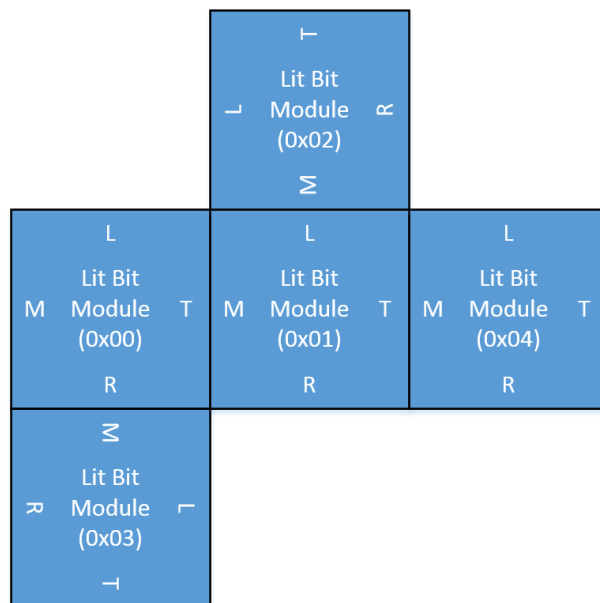


Figure 6: Lit Bit Module Orientation

3.3.3 Specifications

Through the use of the addressing algorithm the ECD is able to have an array that describes the inter-connectivity of all LBMs and their orientation with respect to other LBMs. This allows the ECD to have all of the information necessary to send out subsets of the animation to the appropriate address and also know the orientation of that the subset of visual data.

3.3.4 Alternatives

The alternative designs considered are other implementations of I2C that require that the new LBM on the bus notify the master of the bus. This initiates the master to begin enumerating through the slave addresses (0x01 to 0x7F), looking for device IDs that do not respond with a bus acknowledge and give that address to the new LBM. This alternative meets the self-addressing specification but the ECD has no reference as to where this LDM is located with respect to others in the Lit Bit network. This is inadequate for our design as orientation of the actual visual data is important so that the display is continuous across modules. The choice of I2C is due to the fact that each module can communicate on the bus and do not require a hard address or extra hardware as we need each device to be flexible for modular design. I2C offers this, as well as good documentation of the standard. Other standards that are possible are CAN, LIN, and SMBus. These alternatives require more hardware, and/or custom software design to get them to work within the project's specifications.

3.4 External Control Device Software

3.4.1 Requirements

The External Control Device Software must meet the following requirements:

- Take user input (animation) and translate it into data for the Lit Bit Module(s)

3.4.2 External Control Device Software

The final design of the ECD is implemented on a Raspberry Pi computer. I2C control is implemented with Python, specifically using the SMBus library to transfer the data to the LBMs. The addressing algorithm specified in Section 3.3 is the method that it used to acquire and address all of the LBMs in the network so that each has a unique address to receive the animation data. The animations themselves come from the animation renderer discussed in Section 3.2. This is selected using the user interface by the user and is sent to the python application. The python application takes the animation and the list of available LBMs and sends a subset of the animation to the appropriate LBM in the network. Figure 7 below shows some simple pseudocode of what happens at a very high-level. The Raspberry Pi takes the data in using file I/O (input/output) which is fed in from the animation renderer. The animation is taken in and sent back out to the LBM. The actual software implementation also accounts for the orientation of the LBM.

```
1 import smbus
2 import time
3
4 # Setup RasPi as the bus master
5 bus = smbus.SMBus(0)
6
7 # Create an array containing the Lit Bit information
8 LBM = []
9
10 for x in LBM.length:
11     for y in range(0,144):
12         bus.write_byte_data(LM[x*3, 0, animationArray[y]])
```

Figure 7: Pseudocode of software used in External Control Device

3.4.3 Specifications

Through the usage of the addressing algorithm's local array, the ECD is able to send subsets of the animation visuals to the appropriate LBM and have it in the right orientation for the user to allow for a seamless display.

3.4.4 Alternatives

The alternatives available for this portion of the design is the C programming language. Many I2C libraries exist for C and C-based languages for the Raspberry Pi. The reason Python is used instead is that there are a lot of existing I2C libraries for Python on the Raspberry Pi that abstract the C libraries and make it much easier to develop for. This allows for more time to use in actual implementation of the algorithm and less time building up libraries.

3.5 Mechanical Enclosure

3.5.1 Introduction

The mechanical enclosure is designed to house all of the electrical components required to fully assemble one LBM to increase safety and usability. The electrical components required for each LBM include:

- One assembled Lit Bit PCB.
- One male power and data connector.
- Three female power and data connectors.

3.5.2 Requirements

The mechanical enclosure for a Lit Bit module must meet the following requirements:

- Must isolate the electrical components inside the Lit Bit module from outside electrical disturbances.
- Must provide enough vertical and horizontal room to encompass all electrical components required for one Lit Bit module.
- Must separate individual pixels on the Lit Bit module to mitigate LED light bleed.
- Must provide light diffusion to make pixels look square.
- All pixels on a module and subsequent modules must be equidistant.

3.5.3 Construction Material Selection

Multiple materials are acceptable for constructing the mechanical enclosure. However, to be able to provide electrical isolation, only materials which act as electrical insulators are acceptable for use as they are able to isolate the internal electrical components from the outside world. Acceptable insulating materials include plastics (ABS, PLA, acrylics, polycarbonates) and wood or wood-like materials.

When discussing ease of manufacturing, ABS and PLA plastics are the best choice given that 3D printers are capable of printing with either of these plastics. Wooden and wood-like materials, on the other hand, require a lot manual effort in order to produce an effective enclosure that fits and is consistent.

3.5.4 Enclosure Sizing

Once assembled, the Lit Bit PCB, as designed, measures 195 mm in length, 195 mm in width, and 12 mm in height. The mechanical enclosure must be sized large enough to house the Lit Bit PCB on all six sides while still providing room for the power and data connectors to be exposed to other LBMs.

Many 3D printers have a maximum printing area of approximately 250 mm × 250 mm × 100 mm. In order to be able to print the mechanical enclosure on any average-sized 3D printer, it is acceptable to set the overall size of the mechanical enclosure to 200 mm × 200 mm × 20 mm.

3.5.5 Mitigating Light Bleed

When designing the Lit Bit PCB, it is noted that with 144 LEDs on each board, the user will notice some light bleed between pixels if no physical separation between LEDs exists. To solve this, a *waffle structure* is used to separate each pixel on the board. The LED sits in the centre of each square and is surrounded by four walls. This prevents most of the light from crossing over into another pixel's square. In Figure 8 an illustration of what the waffle structure looks like is displayed.

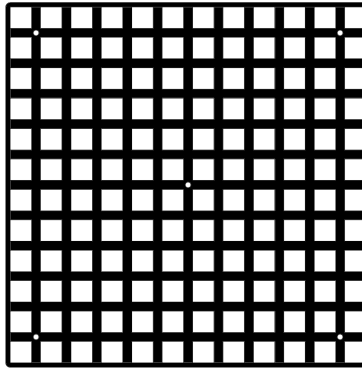


Figure 8: The waffle structure design to prevent light bleed between pixels.

These waffle structure walls are designed to sit directly on top of the Lit Bit PCB to ensure very little light can bleed underneath each wall. With this design, it is evident that light bleed between pixels is mitigated, if not eliminated.

3.5.6 Light Diffusion

Although the waffle structure solves the potential light bleed issue, it does not, on its own, provide any light diffusion properties. Ideally, the effect that is trying to be achieved is when the LED is on, the light emitted by it fills the entire square as shown below in Figure 9.

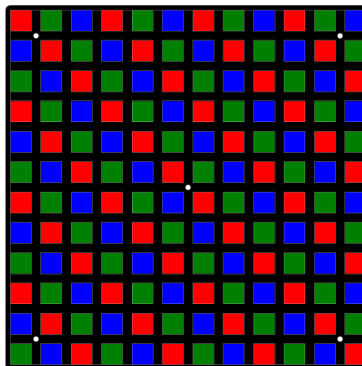


Figure 9: The desired effect of light diffusion.

This effect is able to be achieved using various light diffusing materials. Some acceptable materials include polyurethane foam, light diffusing acrylics, and light diffusing films.

Polyurethane foam must be cut to fit inside the square each LED sits in. This means that 144 square pieces of foam will need to be cut out in order to completely assemble one LBM. Although this is a viable option, it is not necessarily feasible given that multiple LBMs are to be manufactured.

As for light diffusing acrylics and films, these can both be cut as one piece laid over top of the LBM in order to provide the desired light diffusing effect. Thus, for the final design either a light diffusing acrylic sheet or a light diffusing film will be overlaid on top of the LBM.

3.5.7 Equidistant Pixels

To make any configuration of LBMs to act as a seamless display, each pixel of said display must be equidistant. To achieve this, the mechanical enclosure must separate each pixel equally. This is achieved by setting the width of the walls of the waffle structure to be twice the width of the outer walls. When two LBMs are connected, the outer wall thickness effectively doubles and enables pixels between two LBMs to be equidistant. This effect is demonstrated in Figure 10.

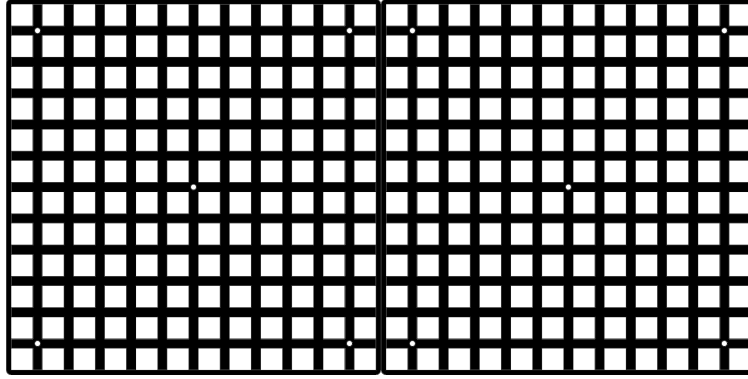


Figure 10: Two LBMs side-by-side with equidistant pixels.

To implement this design, the outer wall thickness is chosen to be 2.5 mm while the waffle structure wall thickness is chosen to be 5 mm.

3.5.8 Proposed Design

The proposed design of the mechanical enclosure is shown in Figure 11 below.

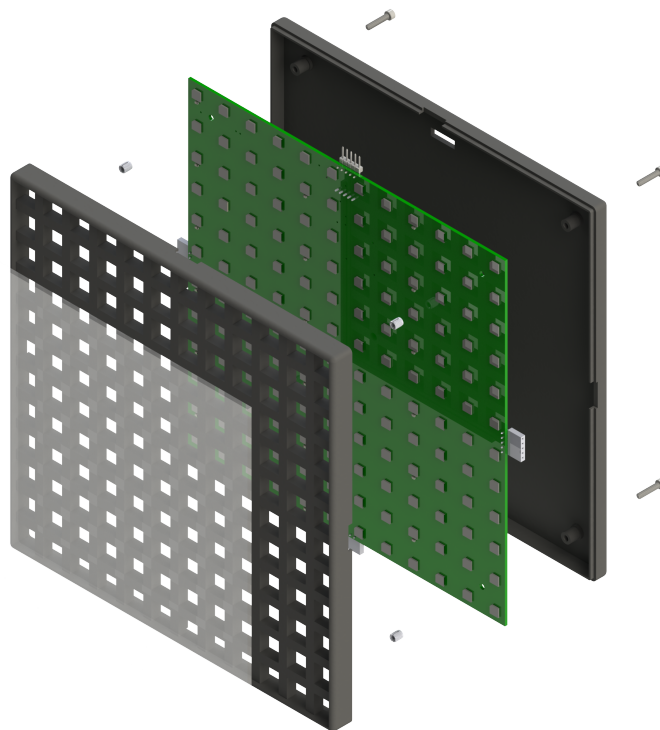


Figure 11: An exploded view of the proposed design for a single LBM.

From Figure 11, one can see that the enclosure consists of three main parts, the top shell, which contains the "waffle structure", the bottom shell which protects the bottom of the circuit board from outside interference, and the light diffusion overlay. The top and bottom shells are secured to each other using socket head cap screws with the Lit Bit PCB sandwiched in between both shells. The light diffusion overlay sits on top of the assembled LBM.

3.6 Mating Connector Integration

3.6.1 Requirements

On each LBM, there exists one male connector and 3 female connectors. This design enables the LBM to have the modularity required to complete the project successfully. These connectors must satisfy the following requirements:

- Must be easy to mate for a user.
- Must only mate in one direction.
- Must be capable of at least 5A of current capacity (refer to Section 3.8).

For this project, a subset of connector types are chosen to compare amongst each other. These connector types are as follows:

- Header pins – Standard 2.54 mm pitch header pins used on a variety of devices; cheap but somewhat unreliable.
- Pogo pins – Spring loaded male pins which make contact with flat pads on the female side; extremely reliable but expensive.
- Blade connectors – These connectors are intended for use with laptop batteries. They have male blades which slide into female conductive *valleys*; reliable but expensive.

In Figure 12, an image of each style of connector is shown.

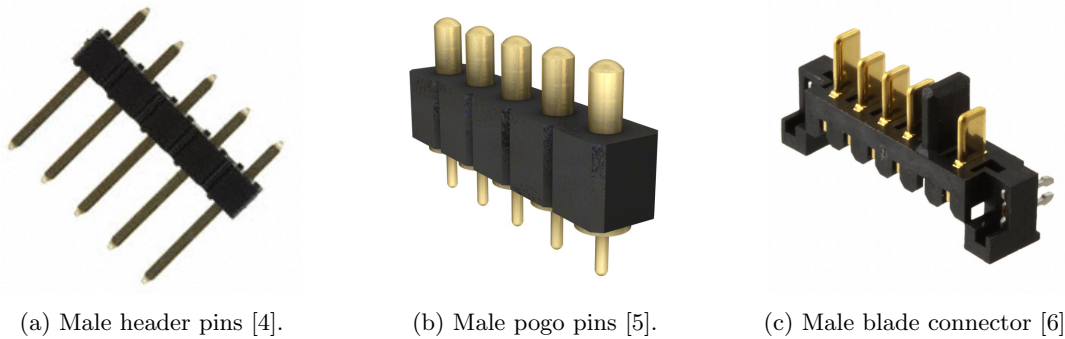


Figure 12: Different connector option for LBMs.

Each of these connectors are compared against one another on each requirement.

3.6.2 Ease of Use

When comparing the three alternative connector types, it is evident that the most easy-to-use connector is the pogo pin style connectors. This is due to them being able to be mated even if the male and female sides come at an awkward angle. Header pins must be mated straight on into the female side with little room for user error. Lastly, blade connectors have the ability to be mated vertically and horizontally which the other two options do not allow. This makes the blade style connectors a second choice.

3.6.3 Mating Direction

Mating direction is a very important issue as it could mean connecting two LBMs backwards which enables a short directly from the 12 V pin to ground.

Header pins on their own do not allow for only a single mating direction. This means that some form of directional safety would need to be designed in order to use header pin connectors. Pogo pins also have this same kind of issue. Blade style connectors, on the other hand, have a plastic block which enables only mating in one direction.

For this project, blade style connectors are the first choice when considering mating direction.

3.6.4 Current Capacity

Standard sized header pins can transfer approximately 3 A of current [4], making them unsuitable for the current rating requirement. Pogo pins can be sized up to continuous currents of 9 A [5], while blade style connectors are typically rated to 6 A of current [6].

Thus, both pogo pins and blade style connectors are suitable for use on the power distribution side of the project. Header pins are not suitable at their standard size of 2.54 mm.

3.6.5 Results

For ease of use, pogo pins are obviously the best choice as they allow for odd mating angles in comparison to header pins and blade style connectors. When comparing on mating direction, only the blade style connectors (by default) have the ability to be mated in only one direction, horizontally or vertically, due to their design. Lastly, both the pogo pins and the blade style connectors have the current capacity required for power distribution in this project.

Therefore, as the overall best solution for the mating connector, the blade style connectors are used for power distribution and data transmission as they best satisfy all three categories.

3.7 Lit Bit Module Microcontroller Software

The final design of the LBM microcontroller software encompasses various subsections of functionality:

- Bus interactions.
- Decoding bus packets for LED use.

The following sections go into finite details about all of these sections.

3.7.1 Bus Interactions

The bus interactions portion of the Lit Bit Microcontroller Software design encompasses all bus transactions which can be broken down into 2 separate buses, the Inter-Integrated Circuit (I2C) bus and the Serial Position Address Line (SPAL).

The I2C bus handles all data transactions that occur between devices. Primarily, transactions are started by the External Control Device (ECD) and then received by the LBMs. Each LBM has it's own unique address assigned to it by the ECD. Data is received at the LBMs in the form of a hexadecimal colour consisting of 8 bits per RGB colour, for a total of 24 bits per LED. However, downscaling is used to minimize the amount of packets sent on I2C bus as well as maximize the number of colour steps for each of the red, green, and blue LEDs. Therefore, a bitmap from 24 bits down to 16 bits is used (refer to Section 3.7.2). Each module

therefore requires a set amount of I2C transactions stipulated in Equation 3.

$$16 \text{ bits specify a colour per LED} = 21\,845 \text{ colours per RGB colour} \quad (1)$$

$$\frac{16 \text{ bits per LED}}{8 \text{ bits per I2C transaction}} = 2 \text{ I2C transactions per LED} \quad (2)$$

$$(144 \text{ LEDs per LBM}) \times (2 \text{ I2C transactions per LED}) = 288 \text{ I2C transactions per LBM} \quad (3)$$

Therefore, 288 transactions per LBM are required on the I2C bus. In order to accommodate this number of transactions per LBM and to satisfy Functional Specification #4, various I2C bus speeds [3] are utilized for the calculation, including *Standard Mode* which runs at 100 kbit/s, *Full Speed Mode* which runs at 400 kbit/s, *Fast Mode* which runs at 1 Mbit/s, and *High Speed Mode* which runs at 3.2 Mbit/s. Table 3 shows various calculations for a single LBM as well as theoretical limits of LBMs running at each I2C speed mode. Note that these calculations are made assuming that as soon as the LED colour packets arrive at the LBM, they are decoded and sent to the LEDs immediately.

Table 3: Comparison of various I2C bus speeds for transmission of data.

Single Lit Bit Calculations					
I2C Speed Mode	Speed (bit/s)	# of Bits per LBM (bit/LBM)	Seconds per LBM (s)	Maximum FPS (FPS)	Theoretical Max. Connected LBMs
Standard Mode	100 000	2304	0.023 04	43.4	1
Full Speed Mode	400 000	2304	0.005 76	173.61	7
Fast Mode	1 000 000	2304	0.002 304	434.03	18
High Speed Mode	3 200 000	2304	0.000 72	1388.89	57

$$\text{Seconds per LBM} = \frac{2304}{\text{Speed}} \quad (4)$$

$$\text{Maximum FPS} = \frac{1}{\text{Seconds per LBM}} \quad (5)$$

$$\text{Theoretical Maximum Number of LBM Connected} = \left\lfloor \frac{\text{Speed}}{24 \times 2304} \right\rfloor \quad (6)$$

As shown in Table 3, when I2C *High Speed Mode* is utilized, a maximum of 57 devices can theoretically be used. However, in reality, long lengths of microstrip traces prove to be susceptible to noise at higher frequencies. Calculations are conducted using Equation 7.

$$\begin{aligned}\text{Length of I2C Line Through 1 LBM} &= 100 \text{ mm} + 200 \text{ mm} + 200 \text{ mm} + 100 \text{ mm} \\ &= 600 \text{ mm}\end{aligned}\tag{7}$$

Thus, 57 modules is far beyond the total number of supported devices for the LBMs due to power distribution limitations (see Section 3.8) and would be highly susceptible to attenuation due to the total microstrip length of the I2C bus line (Equation 7). The second restriction put in place is the selected microcontroller, the *STM32F0*, which has hardware architecture that only supports speeds up to *Fast Mode*. Therefore, I2C *Fast Mode* is used.

The other interactive serial bus features on the LBMs is the Serial Position Address Line (SPAL). The SPAL delegates position tracking between two modules in order to properly address a device as well as position within a chain of LBMs. The SPAL works on the principle of hardware interrupts in adjacent LBMs. When a new device is connected to an existing chain of LBMs, it pulls the SPAL line low, notifying the device it was connected to. The rest of the addressing procedure is then handled by the ECD via the I2C bus.

3.7.2 Decoding Bus Packets for LED Use

Once data is received via the I2C bus into the LBM, the processor must then convert it into a suitable character array that the individual LEDs can understand.

The selected LEDs are the WS2812B which are daisy chained, serially addressable devices [7]. Each LED uses 8 bits for each RGB colour – 8 bits for red, 8 bits for green, and 8 bits for blue – making a grand total of 24 colour bits per LED. Since there are 144 LEDs in each LBM, a grand total of 3456 colour bits are required per LBM.

Downscaling and the use of a bitmap intended to cut down the number of transactions on the I2C bus in order to maximize the FPS on the final output as it is unnecessary for each coloured pixel to have the capability of 0x00 - 0xFF colour steps. Downscaling options include 16-bit and 12-bit downscaling from the ECD to the individual LBMs.

12-bit downscaling made it easy to decompose as $\frac{12}{3} = 4$ bits per colour, meaning abstraction could be implemented by simply padding logical 0s into the least significant bits of each colour, also known as *bit stuffing*. For example, colours are limited to 0 - F (4 bits) where the new colour would be 00 - F0 (8 bits). This makes it very easy to decompose at the processor level and send the data to the individual LEDs.

There are three options available for 16-bit downscaling: 16 bits mapped to addresses in a lookup table (LUT), mapping red, green, and blue to 5 bits, 6 bits, and 5 bits, respectively, or lastly mapping red, green, and blue to 5 bits, 5 bits, and 5 bits with 1 bit of padding, respectively.

The first option, 16 bits mapped to addressed in a LUT, is by far the most complex and resource-heavy implementation as it requires a table that sits in memory with a mapped value for every one of the possible 16 bits. Equation 8 extrapolates on how many different colours are required to be hard mapped to a 16 bit address. All 65 536 positions would then have to be stored in flash memory on the microcontroller and be hard-mapped to key RGB mix values, making this method very inefficient.

$$2^{16} = 65\,536 \text{ colours}\tag{8}$$

The second option, mapping RGB to 5, 6, and 5 bits respectively, is very similar to the last option, mapping RGB to 5, 5, and 5 bits respectively, and will be evaluated fairly similarly. These methods are the easiest to decompose and send to the LEDs as structuring would consist of bit stuffing 0s into the lowest significant bits of each one of the colours received from the bus. For example, a 5-bit colour would be received as `0bXXXXX`, where Xs represent a logical bit that colour is to be represented with, and would be upscaled to 8 bits by shifting 3 0s left into the received 5 bits as follows: `0bXXXXX000`. This provides the correct colour stepping as the LEDs are still capable of going from (0x00) to almost maximum (0xF8).

16-bit downscaling is chosen as each transaction over I2C requires a maximum of 8 bits so two transactions are required in either case. 16-bit is also chosen because many of the available libraries for bit mapping are available for utilization over 12-bit mapping. The chosen 16-bit downscaling method used is option 2 – mapping RGB to 5 bits, 6 bits, and 5 bits respectively – as it enables for more depth in human-perceived colours as green is more receptive to the human eye [8].

Using this 16-bit downscaling method in conjunction with I2C *Fast Mode* satisfies Functional Specification #4.

3.8 LED Array Circuit

3.8.1 Requirements

The design of the LED Array Circuit must be able to achieve the following requirements:

- Minimize current draw through any connected chain of LBMs.
- Step the input voltage down to the LED working voltage (approximately 3 V).
- Accomplish the desired 12×12 array of LEDs required to complete an LBM.

The final design of the LED array circuit must minimize current draw through a given chain of LBMs. The Lit Bits feature a 12×12 grid of LEDs making the total length of the chain 144 LEDs.

3.8.2 Power Calculations

Various alternatives were considered when constructing the LED circuit with power and propagation delay in mind in order to meet non-technical requirement 3, which specifies current draw should be kept to a minimum, and technical requirement 4, which specifies a minimum frame rate of 24 FPS, respectively.

The first metric, power, can be calculated directly from specifications in the WS2812Bs datasheet [7] with current metrics pulled from the WS2812Bs main distributor, Adafruit Industries [9]. Each WS2812B has a current draw of about 60 mA in the extreme case, when the red, green, and blue portions are forward biased at a constant voltage. All calculations are performed with the worst case currents in mind. Equation 9 extrapolates on the current draw of one LBM. Equation 10 extrapolates the voltage across the series chain of LEDs with a voltage spec of 3.5 V.

$$I = 60 \text{ mA} \times \frac{\# \text{ of LEDs in Series}}{144 \text{ LEDs}} \quad (9)$$

$$V = 3.5 \text{ V} \times (\# \text{ of LEDs in Series}) \quad (10)$$

The voltage range of the LEDs is 3.5 V to 5 V, but 3.5 V is chosen as the logical effort to push an individual bit to the LED itself is similar to that of what a standard microcontroller can output (3.3 V). In order to trigger a rising edge in the device itself (WS2812B LEDs) from the microcontroller, a signal of at least 70% of the supply voltage of the LEDs must be provided to them [9]. Equation 11 shows this calculation for the maximum voltage that can be utilized without the use of a logical shifter.

$$\begin{aligned} V_{\max} &= (100\% + 70\%) \times V_{\text{sig}} \\ V_{\max} &= 170\% \times 3.3 \text{ V} \\ V_{\max} &= 5.61 \text{ V} \end{aligned} \quad (11)$$

If the current and voltage values are then extrapolated into various lengths of LEDs in series, the results of Table 4 show the difference in the mentioned specs. Note that the maximum threshold of 1 connector is 6A.

Table 4: Various numbers of LEDs in series to compare voltage and current specs

# of LEDs in Series	Current Multiplier	Total Current (A)	Total Voltage (V)	Max # of LBMs in Series Chain
1	144	8.64	3.5	0
4	36	2.16	14	2
8	18	1.08	28	5
16	9	0.54	56	11

The specification of 4 LEDs in series is chosen for the first design iteration as it incorporates a reasonable amount of LBMs as well as a reasonable power supply voltage. Comparatively, 1 LED in series would not work at all as it exceeds the connector current specification, discussed in Section 3.6. Furthermore, if 16 LEDs in series are used, the project would be deemed "high voltage" and is then unsafe for use in a modular design specification (refer to Specifications 2 and 3 in Section 2.1). That said, the use of 4 or 8 LEDs in series are both close to being within the tolerable safe range of voltages, as well as reasonable amounts of current within the connector specifications.

Generally, these LEDs are likely to be run at approximately 70% of their full brightness to make the display easier on the user's eyes. Under this assumption, 4 and 8 LEDs in series options are further broken down in Table 5.

Table 5: Various numbers of LEDs in series to compare voltage and current specs at 70% current draw

# of LEDs in Series	Current Multiplier	Total Current (A)	Total Voltage (V)	Max # of LBMs in Series Chain
4	36	1.512	14	3
8	18	0.756	28	7

Through this iterative process design, 4 LEDs connected in series is selected as the preferable LED array circuit.

4 Section: Discussion and Project Timeline

4.1 Evaluation of Final Design

The user interface meets the minimum specifications for the project to function whereas the UI translates settings selections to the ECD from the user (Specification 6 in Section 2.1). There may be other functionality that could be added to the UI such as display brightness adjustment or animation speed.

The current design of the Animation Renderer does not effectively accommodate the 24 FPS framerate (Specification x in Section 2.1) as the data structure currently only holds a single animation frame as opposed to all 24 frames.

The final design of the Addressing Algorithm meets all specifications as it is able to automatically provide an address for all of the LBMs in a connected network.

The current design of the External Control Device Software does not effectively accommodate the downscaling of the colour data of individual pixels on the bus and will need some further optimizations for it to be able to be used as a seamless display.

The Mechanical Enclosure, as designed, meets the absolute minimum requirements (Specification 2 in Section 2.1).

The Mating Connector Integration, as designed, helps choose the best overall connector for the project. Theoretically, all three options chosen meet the specifications required for the project (Specification 3 in Section 2.1).

All specifications are met for the LBM Microcontroller Software where a substantial amount of LBMs can be connected using the chosen bus structure. In order to satisfy this objective, the I2C bus needs to be implemented in hardware and robustly tested.

The current LED Array Circuit design does not effectively accommodate for multiple units as current draw is the largest limiting factor through the connectors in this modular design. Future iterations may include the use of more connectors to promote ample current transfer between devices.

4.2 Use of Advanced Knowledge

Both the Addressing Algorithm and External Control Device Software make use of Embedded Software (ECE 455) principles as the software design requires the implementation of embedded C code on the LBMs and working knowledge of the I2C protocol to automatically address each LBM in the network. The LED Array Circuit utilizes Integrated Digital Electronics (ECE 445) knowledge with regards to electrical effort and low to high transition thresholds for sending a logical 1 to a daisy-chained, serially-addressable LED without the use of a logic level shifter. For the User Interface and software communications, prepared statements knowledge, discussed in Database Systems (ECE 356) are used to prevent SQL injection from user input.

4.3 Creativity, Novelty, and Elegance

This design incorporates a sleek, elegant, and simple product that is easily replicated, which cuts down on costs and manual labour required to assemble a single LBM. Lit Bits' modular design allows for users to engage with the device in any way they can imagine, while still maintaining a seamless visual display. This device is simple to use and is the very definition of plug-and-play.

4.4 Student Hours

Table 6: Total student hours from May 1, 2015 to July 2, 2015.

Group Member	# of Hours
██████████	87
██████████	62
██████████	101
██████████	35

4.5 Potential Safety Hazards

4.5.1 High Current

Section 3.8 delves into the utilization of potentially high currents at low voltages that meet the requirements of being within a tolerable range. With high currents, there is a large risk of someone being shocked. It is commonly known that as little as 100 mA can kill a human being, thus caution has to be taken when dealing with any of these circuits. Design precautions are taken to ensure a user cannot touch or short live connections. Furthermore, V_{dd} and V_{ss} rails are separated in order to prevent easy shorting of the power rails.

4.5.2 Eye Safety

The bright LEDs prove to be an eyesight hazard and can cause "welder's eye", damaged vision due to prolonged exposure to bright light. Proper eye wear should be worn when working with the raw, non-diffused LED light.

4.5.3 Heat and Fire Safety

Heat and fire safety prevents issues with both the tools (soldering irons) we use to manufacture a LBM PCB, as well as components that can potentially catch fire from being exposed to a higher currents than the component can tolerate, such as diodes, power transformers, inductors, and other various heavily-loaded equipment. Preventative measures, such as utilizing large ground plane areas on the PCB as heat diffusion, or strategic component placement, fusing critical points, has been put into effect to prevent components from catching fire or overheating. Soldering safety procedures are also put into place, such as turning off the iron when not in use and handling all manufacturing tools with care.

4.6 Project Timeline

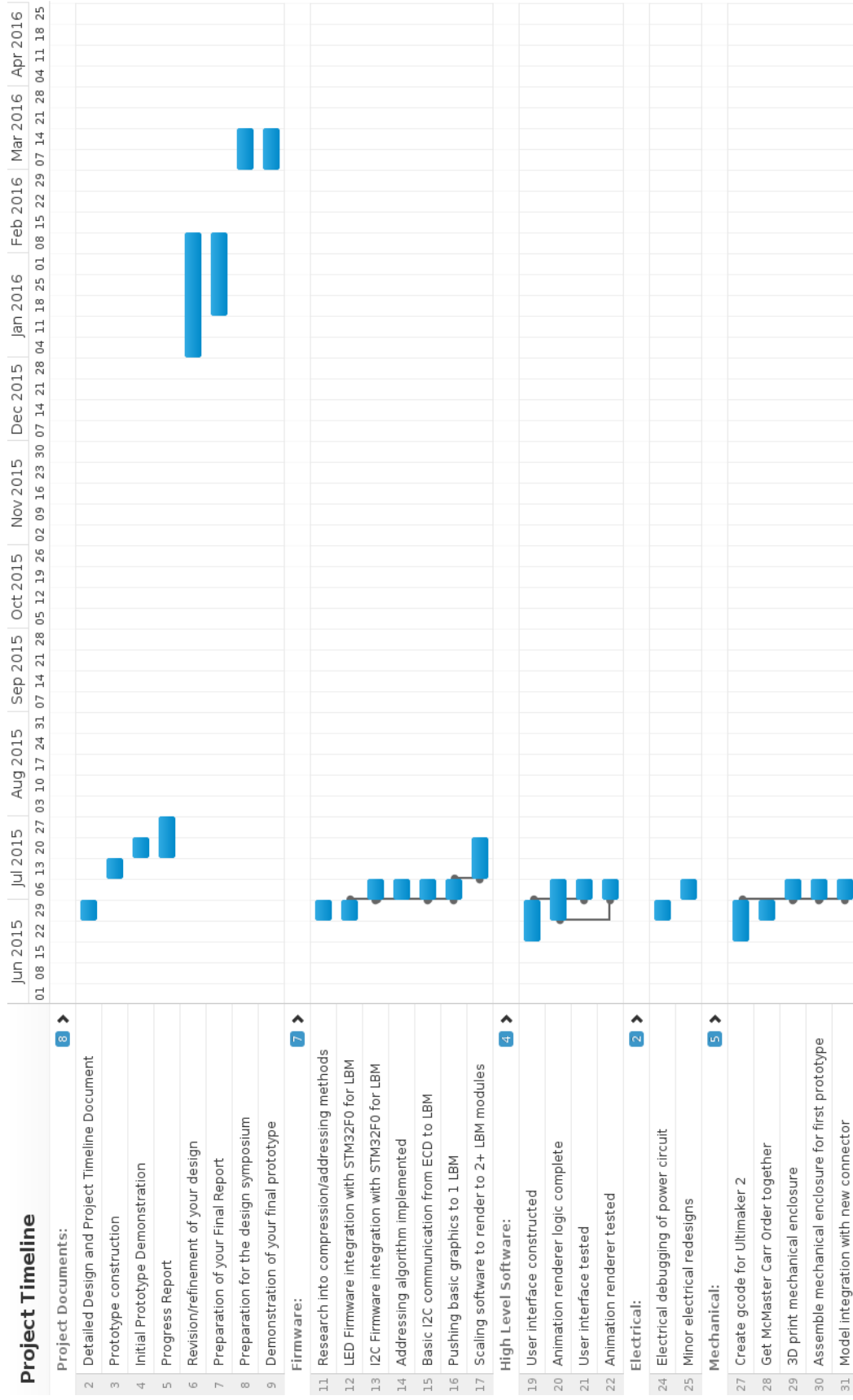


Figure 13: Group 2016.039 work schedule timeline from July 2, 2015 to March 22, 2016.

5 Glossary

ABS – Acrylonitrile Butadiene Styrene; A type of plastic commonly used in 3D printers and tubing.

Apache – web server software

CAN – Controller area network; is a vehicle bus standard designed to allow microcontrollers and devices to communicate with each other in applications without a host computer.

ECD – External Control Device; Refers to the device which controls a connected chain of LBMs.

FPS – Frames Per Second; A measurement of the frequency of images drawn on a display.

HTML – HyperText Markup Language; is the standard markup language used to create web pages.

I2C – Inter-Integrated Circuit; Communication bus invented by Philips Semiconductor used to connect integrated circuits (ICs) to microcontrollers.

LAMP – Linux Apache MySQL PHP; A software package for UNIX-based systems which consists of multiple web tools.

LBM – Lit Bit Module; Refers to one assembled Lit Bit.

LED – Light-Emitting Diode; A p-n junction diode which emits light when it is on.

LIN – Local Interconnect Network; is a serial network protocol used for communication between components in vehicles.

Linux – is a Unix-like and mostly POSIX-compliant[7] computer operating system.

LUT – Lookup Table; An addressed table of values used to quickly convert an input to it's matched output.

MySQL – Structured Query Language; relational database management system.

PCB – Printed Circuit Board; Electrically connects and mechanically supports electronic components using conductive traces.

PHP – Is a server-side scripting language and designed for web development

PLA – Polylactic Acid; A type of biodegradable plastic commonly used in 3D printers.

Prepared Statements – A prepared statement is a feature used to execute the same (or similar) SQL statements repeatedly with high efficiency.

RGB – Red Green Blue; A colour scheme which defines the amount of red, green, and blue exists in a given colour.

SMBus – System Management Bus; is a single-ended simple two-wire bus for the purpose of lightweight communication, it is derived from I2C.

SPAL – Serial Position Address Line; Line used to notified adjacent Lit Bit module of a new connection and position relative to the adjacent Lit Bit Module, it is pulled down to cause an interrupt on the adjacent Lit Bit module

UI – User Interface; The interface in which the user interacts with the system.

WS2182B – An intelligent type of LED which communicates using a specialized serial-based protocol to display different RGB values.

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